

POWER GENERATION SOLUTIONS

Accelerating Time to PowerTM

From the generation asset to the grid interface. Cable products, application support, installation planning and project execution, connected to the plant's electrical scope.

START A CABLE PACKAGE REVIEW

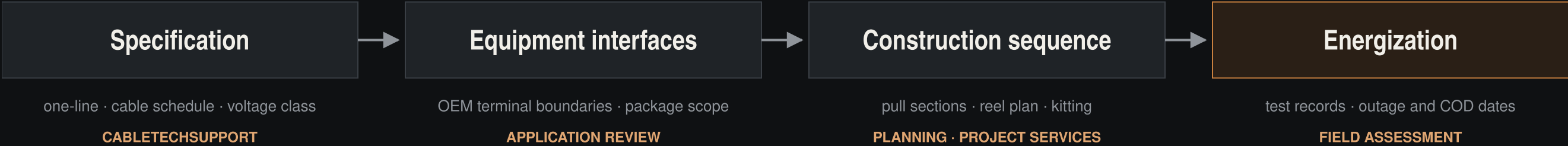
16 application zones / 29 catalog references / 6 documented cases / interactive e-brochure and technical appendix enclosed

01 / THE ELECTRICAL EXECUTION CHALLENGE

Cable is last on every branch and first on the schedule.

Cable selection, equipment interfaces, construction sequencing and energization are one chain. A spec frozen before the interfaces are agreed, a reel out of sequence, or a drive output on the wrong cable each costs time to power.

Time to power Reach the specification before it is frozen, with the plant model, the one-line and the cable schedule in the same conversation. Capability: CableTechSupport application review; Construction Planning Services takeoffs and pull calculations.	Installation labor and constructability Reel lengths, pull sections, circuit identification and staged issue planned around the installation sequence. Capability: Contractor Solutions and SIMpull equipment planning; qualified prefabrication to an agreed engineered scope.	Material availability, coordination and traceability Published constructions with specification numbers, kitting and labels by circuit, reel tracking and staged delivery. Capability: SPEED Services, Project Services, Southwire Reel Tracking System.	Reliable operation and lifecycle support Constructions rated for the duty, and assessment or rejuvenation of existing feeders where a brownfield expansion depends on them. Capability: Field Assessment Services and cable rejuvenation, with published utility results.
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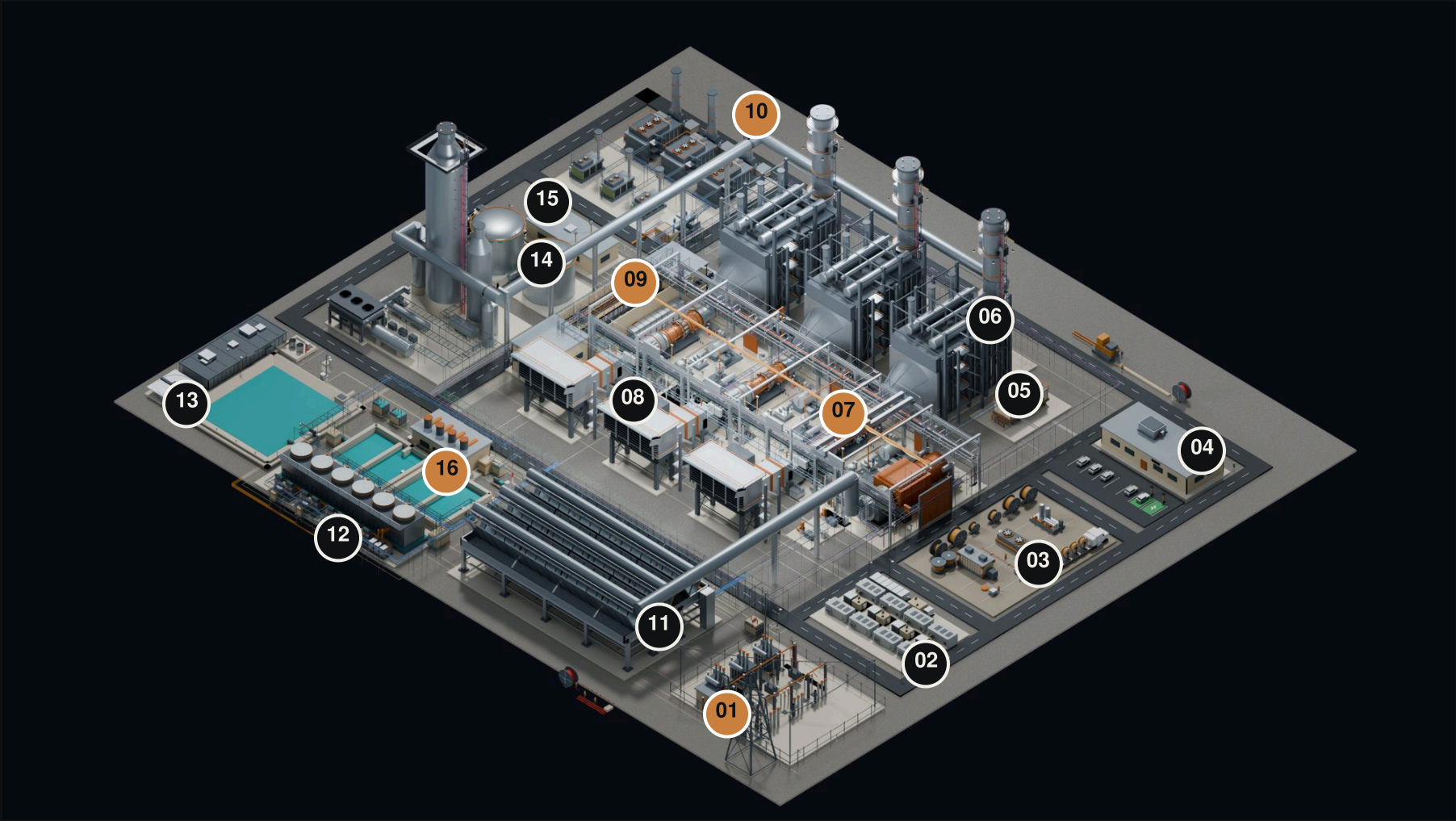


Audience: EPC engineers and procurement teams; generation owners, utilities and independent power producers; OEMs, modular-power companies and equipment packagers.

02 / EXPLORE THE PLANT

One plant. Sixteen application zones.

Every zone is an equipment package, a cable scope and a buyer. The interactive e-brochure opens each one with its cable families, products, services and specification links. Copper pins mark the zones most conversations start in.



- 01 Switchyard / GSU / grid tie
- 02 Battery energy storage
- 03 Reel staging / prefab
- 04 Admin / control room
- 05 E-house
- 06 HRSG trains & stacks
- 07 Turbine hall
- 08 GT air inlets
- 09 MCC / VFD / UPS
- 10 Modular power yard
- 11 Air-cooled condenser
- 12 Chillers / cooling towers
- 13 Water / wastewater
- 14 Carbon capture
- 15 Gas metering
- 16 Shared cable corridor

Concept application map from the original engineering model. Optional systems are shown. Illustrative, not a validated engineering design or digital twin; cable design and quantities follow the approved project scope.

02 / EXPLORE THE PLANT

From equipment to scope: the six zones that set the package.

The handoff, the primary distribution, the core, the drives, the modular yard and the corridor that connects them. The other ten zones are in the technical appendix and the e-brochure.

01 Switchyard / GSU / grid tie

Station service and protection cable package

LV station service; CT/VT and breaker-control cable; trip/close DC; protection fiber; ground grid and equipment bonds.

Package focus: Define utility and GSU boundaries. HV underground cable is a separate engineered scope.

Specifies & buys: EPC / substation contractor via distribution; owner standards and utility team influence.

REFERENCES [59563399](#) [59563699](#) [59547599](#) [59945899](#)

07 Turbine hall

Turbine auxiliary and critical-power package

MV starting and pump feeds; LV lube-oil, turning-gear and vacuum loads; DLO emergency DC; control and temperature signals.

Package focus: Respect turbine OEM interfaces. Generator isolated-phase bus is separate.

Specifies & buys: Turbine OEM defines internal boundaries; EPC / electrical contractor buys external auxiliary circuits.

REFERENCES [167022R108 / 167024R108](#) [59212899](#)

10 Modular power yard

Repeat genset, black-start and fuel-cell interfaces

MV collectors; LV emergency and auxiliary power; OEM-approved starting-battery or fuel-cell DC; synchronization, control and data.

Package focus: Build repeat genset / RICE / fuel-cell interface kits from released drawings.

Specifies & buys: Genset / RICE / fuel-cell OEM and integrator for internal wiring; EPC for yard interconnects.

REFERENCES [167022R108 / 167024R108](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage.

05 E-house

Primary distribution and factory interface package

5 kV or 15 kV-class MV source/tie feeders; Royal Guard LV secondaries; control, protection, DC, data and bonding.

Package focus: MV examples: 55052299 (5/8 kV) and 55050899 (15 kV). Confirm the one-line and separate OEM internal wiring from EPC field connections.

Specifies & buys: E-house / switchgear OEM for internal wiring; EPC / distributor for field feeders and ties.

REFERENCES [55052299](#) [58950101 / 58950104](#) [55050899](#) [188040509](#)

09 MCC / VFD / UPS

Motor-control and essential-power package

LV distribution; shielded VFD motor cable and terminations; UPS input/output and bypass; DLO battery feeds; control and bonding.

Package focus: Match output cable to the drive. Identify redundant AC and DC circuits.

Specifies & buys: MCC / VFD / UPS OEM for internal assembly; EPC and electrical contractor for field terminations.

REFERENCES [64346799](#) [167022R108 / 167024R108](#) [59193199](#) [188040509](#)

16 Shared cable corridor

Plant-wide feeder and routing work packages

MV, LV, control, instrumentation, essential power, fiber and grounding routes connecting the plant zones.

Package focus: Create pull sections and reel plans. Count each circuit once. Select 5/8, 15 or 35 kV constructions from the actual system voltage and routing design.

Specifies & buys: EPC electrical / routing engineer, construction team and distributor; owner for resilience criteria.

REFERENCES [55052299](#) [55050899](#) [89008499](#) [67747299](#)

03 / THE CABLE OFFERING

Seven families cover the plant's electrical scope.

Published references connect circuit requirements to specific constructions. Stock numbers do not indicate inventory. The full catalog with specifications and documents is in the technical appendix.

Medium-voltage power

5/8 kV, 15 kV and 35 kV MV-105 constructions for plant feeders, source and tie circuits, collectors and transformer feeds. Select the construction from the actual system voltage, insulation level and routing design.

[55052299](#) [55050899](#) [89008499](#) [67747299](#)
[66131915](#) [89066999](#)

Low-voltage distribution and motor feeders

600 V and 600/1000 V EPR-insulated singles and multiconductor tray-rated feeders for secondaries, motors, panels and equipment.

[58950101 / 58950104](#) [58949999](#) [59199899](#)

VFD cable

Shielded drive-output constructions, flexible or armored, matched to the drive system. Qualify the drive before selecting the cable.

[45834799](#) [64346799](#) [89052099](#)

Control and instrumentation

CT/VT and breaker-control multiconductors, shielded pairs and triads for analog, status, RTD and thermocouple circuits, and armored instrumentation for classified areas. Type TC and TC-ER are distinct ratings.

[59563399](#) [59563699](#) [59547599](#) [59193199](#)
[577883 \(base code\)](#) [59212899](#) [56295201](#)
[89057289 / 89057399](#) [59945899](#) [59548199](#)

Flexible, portable and approved DC applications

DLO flexible power for emergency and battery circuits, and RenewaFLEX for approved internal BESS wiring. Fixed wiring and portable cable are different applications; BESS DC is not solar PV.

[167022R108 / 167024R108](#) [13892403](#)

Grounding and bonding

Bare copper and insulated equipment grounds for the ground grid, equipment bonds and shield grounds. Final sizes follow the grounding study.

[10704542 / 16266909](#)

Communications, building systems and specialty

Fiber and industrial networks, building-system and branch wiring, switchboard wire, terminations and accessories, and OEM or special-environment leads that follow equipment specifications.

[188040509](#) [SIMpull Barrel THHN/THWN-2](#) [Genesis](#)
[VFD spec sheets 85451 / 85452 / 85456](#)

OEM internal wiring vs. field interconnection. The OEM wires inside the package to terminal boxes or local panels; the EPC or installer connects between packages.

VFD duty. Drive-output cable is matched to the drive system; qualify the drive before selecting the cable.

Fixed wiring vs. portable cable. DLO and other flexible constructions are portable-cable applications, not substitutes for fixed tray-rated wiring.

Hazardous locations. ARMOR-X MC-HL constructions, glands and seals follow the area classification and the complete installation.

MV insulation level. Voltage class, 100% or 133% insulation level and shield type are project selections from the one-line.

BESS DC vs. solar PV. RenewaFLEX is approved internal BESS wiring, not a photovoltaic application. Immersion and circuit integrity are separate qualifications.

04 / FROM SPECIFICATION TO INSTALLATION

Make the schedule work in the field.

Coordinate cable, equipment interfaces and the installation sequence before material reaches the jobsite. Package the delivery around the pull sequence: reel lengths, circuit identification, staging.

PLAN & PREPARE	Construction Planning Services Takeoffs, BOMs, field drawings, coordination and pull calculations. Qualify circuit marking, pull heads and bundling against the installation sequence. Construction Planning Services Service overview · PDF
SPECIFY	CableTechSupport Application review for construction and application questions before the specification freezes. CableTechSupport
PROCURE	SPEED Services Qualified custom requests and schedule needs. SPEED Services
DELIVER	Project Services Coordination, marking, kitting, labels and bundles packaged around the installation sequence. Project Services
DELIVER	Reel tracking Qualify reel tracking, pull lengths and staged issue with the project team. Southwire Reel Tracking System
INSTALL	Contractor Solutions & SIMpull Bring the field team into route layouts, pull calculations and equipment setup. Align handling, pulling and crew preparation before cable arrives. Contractor Solutions SIMpull Cable-in-Conduit
ASSESS & SUSTAIN	Field Assessment & rejuvenation Evaluate existing underground systems with Field Assessment Services. For eligible aging cable, assess rejuvenation alongside repair or replacement and the planned outage window. Field Assessment Services Cable rejuvenation



Reel staging and prefab, zone 03. Staging adds no circuit footage; it changes when and how the footage is issued.

Prefabricated cable spine, project development. A factory-built spine needs defined interfaces, a qualified assembly provider and an agreed test plan. Evaluate a pilot using factory labor, field labor, logistics, schedule and total installed cost. Availability and performance are project-specific.

05 / BEYOND ONE PLANT CONFIGURATION

Build the package around the application.

The combined-cycle model is a starting point. Adapt scope to the generation technology, site and operating duty. Do not combine plant configurations or carry the model's preliminary quantities into procurement.

CONVENTIONAL GENERATION Combined cycle, simple cycle & peakers Plant MV distribution, turbine auxiliaries, HRSG instruments, cooling motors and switchyard protection. A simple-cycle plant has a different balance of plant. Review zone 07 · Turbine hall	MODULAR / BEHIND THE METER RICE, fuel cells & onsite power Separate internal OEM wiring from yard feeders, starting/DC circuits, controls and grounding. For data centers, microgrids and industrial CHP, confirm the generation-to-campus handoff. Review zone 10 · Modular power yard	GENERATION + STORAGE BESS & hybrid sites Distinguish internal battery wiring, field DC interconnects and AC collector feeders. Confirm voltage, polarity, terminals, fire requirements and the integrator's approved cable design. Review zone 02 · Battery energy storage	EXISTING ASSETS / EXPANSION Brownfield upgrades & new process loads Assess aging feeders and outage windows. Add a separate motor and instrument schedule for carbon capture, water treatment or cooling expansion; verify spare route and switchgear capacity. Review zone 14 · Carbon capture
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Generation scope

Generation assets and the electrical balance of plant, up to the generator step-up transformer. Includes behind-the-meter generation packages wherever they sit.

Downstream facility scope

Campus distribution, UPS, PDU and the consuming infrastructure belong to the facility. Confirm the generation-to-campus handoff at the equipment boundary.

Grid interface

Agree the GSU, point of interconnection, asset owner and route before selecting the connection cable. Underground and overhead need different engineering packages.

Underground transmission

Southwire's published 69–500 kV scope includes cable systems, accessories, engineering, installation and testing. Qualify voltage, route, bonding, terminations and the utility interface.

[Products & services](#) [Transmission line card · PDF](#)

Overhead gen-tie & grid upgrades

Compare conventional conductors with C7 composite-core options when thermal capacity, span, sag or reconductoring drives the design. Final selection belongs to the line engineer and utility.

[Conductor comparison · PDF](#)

06 / DOCUMENTED EXPERIENCE

Results with a reference.

Historical Southwire project examples with their published sources. The PowerGen application under each case is a proposed use, not an outcome from the illustrated plant.

CUSTOM MV CABLE / CANADIAN METALS & MINING 15 weeks From RFQ to shipment. CUSTOMER PROBLEM An unusual MV construction and quantity against a 20-week reference lead time. SOUTHWIRE CONTRIBUTION Custom MV cable engineered and manufactured to the project's requirement. PUBLISHED RESULT Southwire reports shipment in 15 weeks against a 20-week reference, with a 33% reduction in minimum order quantity. POWERGEN APPLICATION · PROPOSED Bring cable engineering in early for unusual e-house, ACC or process-feeder constructions and quantities. Read the case study · PDF	FEEDER LIFE EXTENSION / AES OHIO 42,255 ft Rejuvenated at Dayton Airport. CUSTOMER PROBLEM Aging MV feeders with a replacement estimate and a limited outage window. SOUTHWIRE CONTRIBUTION Cable rejuvenation of the existing feeders in place. PUBLISHED RESULT Southwire reports \$530,161 saved against estimated replacement cost, with the feeder project completed in two months. POWERGEN APPLICATION · PROPOSED Assess rejuvenation for suitable aging MV feeders during brownfield life-extension planning. Read the case study · PDF	FIELD INSTALLATION / BJC WEST COUNTY HOSPITAL 16 pulls A coordinated installation setup. CUSTOMER PROBLEM Sixteen feeder pulls of 220–230 feet with large parallel conductors on a working site. SOUTHWIRE CONTRIBUTION Pull calculations, reel setup and coordinated equipment planning with Contractor Solutions and SIMpull equipment. PUBLISHED RESULT Southwire documents 16 pulls of 220–230 feet using 4 × 750 kcmil aluminum conductors with ground. POWERGEN APPLICATION · PROPOSED Use pull calculations, reel setup and coordinated equipment planning for auxiliary feeders. Confirm the actual route and conductor construction. Read the case study · PDF
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More in the resource library: [Onshore HV connection / Vineyard Wind 1 \(32 miles\)](#) · [URD reliability program / NV Energy \(\\$800 million\)](#) · [69 kV transmission / Atlantic Municipal Utilities \(1969 vintage\)](#).

Connect your next power project.

Start with the one-line diagram, the cable schedule, the equipment interfaces and the target energization date.

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START A CABLE PACKAGE REVIEW

Interactive e-brochure: explore the plant, the films and the searchable catalog online. Technical appendix: all sixteen zone packages and the full catalog with specification links.

- 01

Electrical basis
One-line, system voltage, load and fault data, circuit schedule, grounding and redundancy requirements.
- 02

Route & environment
Tray, duct or direct burial; lengths and bends; ambient heat, chemical exposure, wet locations and hazardous-area classification.
- 03

Interfaces & responsibility
OEM terminal boundaries, GSU/POI limits, specification owner, package buyer, installer and testing responsibility.
- 04

Release & documentation
Design-freeze and energization dates, reel and delivery plan, approved substitutions, test records and project-specific origin or environmental documentation.

Based on the September 2026 PowerGen strategy update, the PowerGen email brochure and linked Southwire references. Final engineering and commercial terms are project-specific.